Ericsson PE (HEB-PE-12) Optical Transport Failure: SFP/XFP Degradation, Fiber Cuts, Rx/Tx Power Loss Resolution SOP

Node: HEB-PE-12 | **Vendor:** Ericsson | **Severity:** Minor | **Alarm Code:** ALARM_5220

Description

This Standard Operating Procedure (SOP) addresses the failure domain of Optical Transport Failure on the Ericsson PE (HEB-PE-12) at the Hebbal Edge site. The PE node plays a critical role in the Optical Transport Network (OTN), responsible for transmitting and receiving high-speed data channels over fiber-optic cables. The Ericsson PE employs SFP/XFP transceivers, which can degrade over time due to factors such as temperature fluctuations, optical power levels, and aging. Fiber cuts or breaks can also lead to data packet loss and network downtime. The ALARM_5220 alarm code, with a severity of Minor, indicates a potential issue that requires prompt attention to prevent network degradation or service disruption. This SOP outlines the diagnostic and resolution steps to identify and rectify the root cause of the Optical Transport Failure, ensuring minimal downtime and maintaining network reliability.

Pre-requisite Checks

- Verify that the Hebbal Edge site is accessible and safe for on-site interventions.

- Check the status of redundant links to ensure they are operational and can take over in case of a failure.

- Confirm that the necessary equipment and tools are available for the on-site technician.

- Review the site-specific documentation and previous incident reports to identify potential causes.

Potential Root Causes

Cause	Probability
SFP/XFP degradation due to aging or environmental factors.	High
Fiber cuts or breaks due to physical damage or aging.	Medium
Rx/Tx power loss due to incorrect SFP/XFP settings or issues.	Low

Diagnostic Commands & Expected Outputs

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Command: admin show transceiver, Expected Output: Transceiver status, including temperature, optical power levels, and error counts., Description: This command displays the current status of the SFP/XFP transceiver, including temperature, optical power levels, and error counts. It helps identify potential issues with the transceiver or its connection.
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Command: admin show link, Expected Output: Link status, including transmission power, receive power, and signal-to-noise ratio., Description: This command displays the current status of the fiber link, including transmission power, receive power, and signal-to-noise ratio. It helps identify potential issues with the fiber link or its connection.

Command: admin show alarm, Expected Output: Alarm status, including alarm codes and descriptions., Description: This command displays the current alarm status, including alarm codes and descriptions. It helps identify the root cause of the ALARM_5220 alarm code.
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Resolution Steps

Step 1: Step: Replace the faulty SFP/XFP transceiver., Command: admin replace transceiver slot 1/1/1, Description: Replace the faulty SFP/XFP transceiver with a new one to ensure proper transmission and reception of data packets.

Step 2: Step: Verify and adjust Rx/Tx power levels., Command: admin set transceiver slot 1/1/1 power-level rx 10, Description: Verify and adjust the Rx/Tx power levels to ensure optimal transmission and reception of data packets.

Step 3: Step: Inspect and repair or replace the fiber link if necessary., Command: admin show link, Description: Inspect the fiber link and repair or replace it if necessary to ensure proper transmission and reception of data packets.

Post-Check Verification

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Command: admin show transceiver, Expected Output: Transceiver status, including temperature, optical power levels, and error counts., Description: Verify that the SFP/XFP transceiver is functioning correctly and that the temperature, optical power levels, and error counts are within normal ranges.

Command: admin show link, Expected Output: Link status, including transmission power, receive power, and signal-to-noise ratio., Description: Verify that the fiber link is functioning correctly and that the transmission power, receive power, and signal-to-noise ratio are within normal ranges.

Command: admin show alarm, Expected Output: Alarm status, including alarm codes and descriptions., Description: Verify that the ALARM_5220 alarm code has been cleared and that there are no other active alarms.
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Roll-back Procedures

- Step: Restore the original SFP/XFP transceiver., Command: admin replace transceiver slot 1/1/1 original, Description: Restore the original SFP/XFP transceiver if the replacement transceiver does not resolve the issue.

- Step: Revert Rx/Tx power levels to their original settings., Command: admin set transceiver slot 1/1/1 power-level rx original, Description: Revert the Rx/Tx power levels to their original settings if adjusting them does not resolve the issue.

- Step: Re-inspect and repair or replace the fiber link if necessary., Command: admin show link, Description: Re-inspect the fiber link and repair or replace it if necessary to ensure proper transmission and reception of data packets.